

Programme	AnalystLab Africa Experience Lab Internship Programme
Project	HealthConnect Clinic Experience Lab
	Immunisation Outcomes Assessment Attendance and

324



Felix
3 tabs

Document Control

Version: 1.0

Issue Date: Week 6

Status: Working submission package — technical integration evidence remains subject to verification.

1. Week 5 – Week 6 Transition

The main Week 5 output was the HealthConnect Project Execution & Progress Management structure. Its most important result was the conversion of the Week 4 project foundation into actionable task control, responsibility mapping, priorities, progress monitoring, risks, dependencies and coordination points.

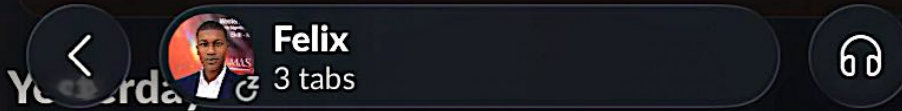
The main limitation carried into Week 6 was the lack of verified technical workstream status within the PM evidence set. During Week 6, a Data Analytics Week 5 analysis was subsequently received and reviewed, creating a verified Data Analytics – Project Management integration point. Other technical workstream outputs remain subject to verification.

Relevant workstreams are Data Analytics, Data Science, Machine Learning Engineering and Generative AI. The Week 6 improvement is to establish how these outputs connect, identify what is ready or still requires development, and manage the conditions required for broader testing.

2. Integration Objective

The Week 6 PM integration objective is to establish a clear, evidence-based view of how HealthConnect workstreams connect, which dependencies affect progress, which deliverables are ready for integration, which activities require further development, what issues and decisions require management attention, and what must be completed before Week 7 testing and refinement.

3. Workstream Integration Map

**Felix** 3:49 pm

Hi Fency 🙋

I'm Felix from the Project Management track, Pod 4, and I'm looking to collaborate with someone from the Data Analytics track for our HealthConnect Week 6 Integration, Advanced Development & Validation activities.

My Week 6 focus is on workstream integration, dependencies, risks, issues, decisions, progress tracking, and preparing the project for Week 7 testing/refinement.

What I'd like to receive from Data Analytics:

- A brief update on the appointment attendance/no-show analysis completed so far.
- Key findings or patterns identified.
- Important factors/features associated with no-shows.



Felix has paused their notifications



Message Felix



Home



DMs



Activity



More



**Felix**

3 tabs



- Important factors/features associated with no-shows.
- Any data limitations, quality issues, or assumptions identified.
- Any dependencies or inputs that may be important for the next stage.
- Any analytical findings that should be considered for Week 7 testing/refinement.

What I can exchange from the Project Management side:

- The HealthConnect project objective and relevant Week 6 integration requirements.
- Information on workstream dependencies and how the analytics output connects to the overall solution.
- Current PM priorities, risks, issues, and unresolved dependencies relevant to the analytics work.
- The Week 7 testing/refinement focus and what outputs need to be ready.
- I can also document how the analytical findings contribute to the overall project and update the relevant



Felix has paused their notifications



Message Felix



Home



DMs



Activity



More



23:34

28%



Felix
3 tabs



Fency 3:51 pm

Hello Felix hope you well , yes I would love to collaborate with you



Felix 3:54 pm

I'm doing well, thank you. I'm glad you're interested in collaborating.

I'll share the specific information I need from the Data Analytics side, and I'll also share the relevant PM information and Week 6 requirements from my side so we can have a useful exchange.

Looking forward to working with you!



From the Data Analytics side, what I mainly need for my Week 6 PM integration work is:

1. Your current analysis/output on appointment attendance and no shows.
2. Key findings: especially any patterns or factors you identified that appear relevant to missed appointments.
3. Important features/variables that you think may be useful for the wider



Felix has paused their notifications



Message Felix



encountered during your analysis.



Home



DMs



Activity

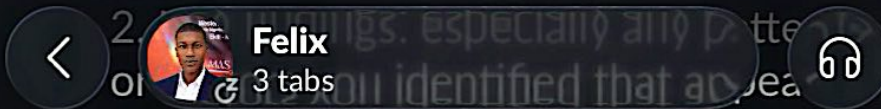


More



23:34

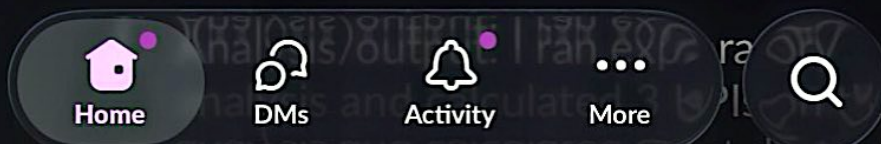
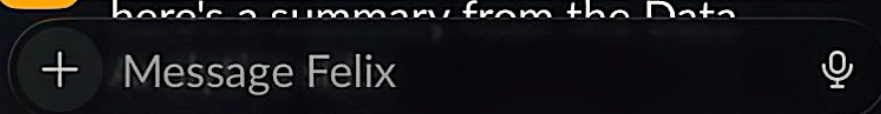
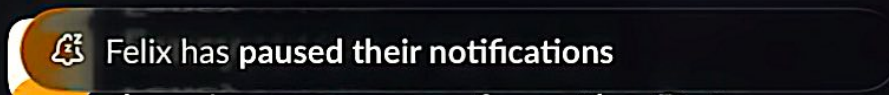
28%



1. Your current analysis/output on appointment attendance and no shows.
2. Findings, especially any patterns you identified that are relevant to missed appointments.
3. Important features/variables that you think may be useful for the wider HealthConnect solution or downstream modelling.
4. Data limitations or issues you encountered during your analysis.
5. Any dependencies you have on other tracks or any inputs you think will be needed for the next stage.
6. Anything from your analysis that you believe should be considered when we move into Week 7 testing and refinement.

I'll use the information to understand and document the Data Analytics = Project Management integration, including how your findings contribute to the overall project objective and whether they require any changes to our dependencies, risks, priorities, or testing preparation.

You can share the analysis file/output or a summary of the findings, whichever is easier.



**Fency** 11:16 pm

here's a summary from the Data Analytics side:

1. Analysis/output: I ran exploratory analysis and calculated 3 KPIs on the HealthConnect appointment dataset (5,000 records) looking at no-show patterns full write-up and Power BI dashboard attached/linked below.

1. Key findings:

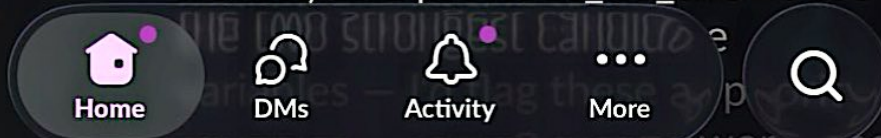
Booking lead time is the strongest predictor of no-shows: patients booking 31+ days ahead no-show at 75.81%, vs. 27.81% for those booking within a week.

A patient's own no-show history is highly predictive: no-show rate climbs steadily from 43.51% (no prior no-shows) to 100% at 5 previous no-shows.

Appointment type has minimal influence (44.64%–51.23% range across all types) much weaker than the two factors above.

Felix has paused their notifications

+ Message Felix





Felix

3 tabs



1. Important features/variables for downstream use: booking_lead_days (or the lead_time_bracket I derived from it) and previous_no_shows are the two strongest candidate variables — I'd flag these as priority features for any modelling work, and as priority risk indicators for scheduling/reminder decisions.

1. Data limitations: distance_to_clinic_km and waiting_time_minutes have some missing values (90 and 60 records respectively, out of 5,000) — not used in my KPIs but would need handling if used downstream. The dataset also doesn't capture *why* patients no-show, only that they did.

1. Dependencies: No blocking dependencies on other tracks for my Week 5/6 work. My main dependency going forward is wanting to know if Data Science ends up using these two variables in their model, so I can validate



Felix has paused their notifications



Message Felix



Home



DMs



Activity



More



23:35

1. For Week 7 testing: worth testing whether the lead-time and no-show-



Felix
3 tabs



(age group, gender, appointment type) before the clinic acts on them that's actually what I'm planning to validate in my own Week 6 work.

Full document + Power BI file: <https://github.com/Fencylebethe/HealthConnect-Week5-Analysis>



HealthConnect_Week5_Analysis

46 KB Word document

1. Review of Week 4 Foundation

In Week 4, the Data Analytics track focused on understanding the HealthConnect appointment dataset and preparing the foundation for analysis. A full data quality assessment was conducted (missing values, duplicates, and category validity checks), three business questions were defined relating to booking lead time, previous no-show history, and appointment type/time slot, and four corresponding KPIs were proposed and justified.

Key resources used were HealthConnect_Appointment_Data.csv and HealthConnect_Data_Dictionary.xlsx. The main assumption carried forward was that reminder_channel values of "None" indicate no reminder was sent rather than missing data. The main limitation identified was missing values in distance_to_clinic_km and waiting_time_minutes (90 and 60 records respectively).

The proposed focus for Week 5 was to calculate the four proposed KPIs, visualise them, and begin exploring relationships between variables to identify the strongest predictors of no-shows. This is the work carried out below.

2. Data Preparation

This Week 5 analysis was conducted using Power BI Desktop, in line with the Data Analytics track's specified toolset (SQL, Python, Power BI, Tableau). The lead_time_bracket column was created using a DAX calculated column, and the No-Show Rate KPI was calculated using a DAX measure (DIVIDE of No-Show count over total appointment count), applied consistently across all three KPI visualisations.

3. Exploratory Data Analysis

Exploratory analysis focused on investigating appointment attendance and no-show patterns across three dimensions identified in Week 4: booking lead time, previous no-show history, and appointment type.

PivotTables were used to cross-tabulate appointment_outcome (Attended, Cancelled, No-Show) against each variable of interest, allowing no-show rates to be calculated as a proportion of total appointments within each category. Across the full dataset, No-Shows (2,423) substantially outnumbered Cancellations (263), indicating that most missed



HealthConnect_Week5_Analysis

211 KB Zip



Felix has paused their notifications



Message Felix



Home



DMs



Activity



More



evidence

23:35

3. Exploratory Data Analysis
Exploratory analysis focused on investigating appointment attendance and no-show patterns across three dimensions identified in Week 4: booking lead time, previous no-show history, and appointment type.

PivotTables were used to cross-tabulate appointment outcome (Attended, Cancelled, No-



Felix
3 tabs



HealthConnect_Week5_Analysis
211 KB Zip



Fency 11:57 pm

Once you've worked this into your PM docs, could you let me know what changed or how it's being used? I need that for my own Week 6 integration evidence



1 reply Today at 10:22 am

Today



Felix 10:33 pm

Thanks again for sharing the Week 5 analysis. I've reviewed it, and here's how I'm using it for my Week 6 PM integration evidence:

What changed / how it is being used:

Your Week 5 analysis is now being used as a documented Data Analytics → Project Management integration input.

Specifically:



Felix has paused their notifications



Message Felix



Home



DMs



Activity



More



23:35

Project Management integration input.



Felix

3 tabs

Specifically.



- The findings on booking lead time and previous no-show history are being recorded as important analytical inputs for the wider HealthConnect solution.
- The finding that appointment type has limited influence is being considered when documenting project priorities and solution considerations.
- The data limitations you identified, including missing distance-to-clinic and waiting-time records, are being captured as limitations/risks that should be considered during later testing and refinement.
- Your identified remaining work, especially the fourth KPI and multi-variable/combined-factor analysis, is being carried forward as an item to consider for the Week 7 testing/refinement stage.
- I'm also updating my integration/dependency records and project coordination plan to show how the Data Analytics output contributes to the overall HealthConnect objective.



Felix has paused their notifications



Message Felix



Home



DMs



Activity



More



**Felix**

3 tabs



is no longer just a Week 5 analytical output; it is now being used as evidence for PM integration, dependency/risk tracking, and Week 7 testing preparation.

I'm not claiming that the findings have already been implemented in the Data Science/ML model, because I don't have verified evidence of that. The PM evidence is specifically the use of your analysis for project integration and coordination.

For your Week 6 evidence, you can describe the exchange as:

Data Analytics → Project Management: Week 5 analytical findings were shared with PM and used to update integration considerations, risks/limitations, remaining work, and Week 7 testing/refinement priorities.

Hope this gives you what you need for your evidence.



Felix has paused their notifications



Message Felix



Home



DMs



Activity



More



23:35

refinement priorities.



Felix

3 tabs

your evidence.



HealthConnect_Week6_PM_Integr...

43 KB Excel spreadsheet

HealthConnect Week 6 — Workstream Integration Map

Shows how workstreams connect and what information/output flows between them.

Integration ID	From Workstream	To Workstream	Input / Output	Why Relevant	PM Coordination Role	
INT-001	Data Analytics	Data Science	Validated analytical findings	Can inform modeling decisions	Track dependency and resulting decision	Valu
INT-002	Data Science	ML Engineering	Candidate model and interface requirements	Enables pipeline integration	Monitor readiness and dependency	Mor
INT-003	ML Engineering	Data Science	Pipeline/model validation feedback	Can reveal compatibility issues	Track issue and resolution	Inte
INT-004	GenAI	Project Management	Assistant evaluation findings, issues and dependencies	Supports readiness and risk decisions	Capture issues, risks and decisions	Eval
INT-005	Project Management	All Workstreams	Priorities, risks, issues, decisions and timelines	Provides project governance	Coordinate integrated controls	Upd
INT-006	All Workstreams	All Workstreams	Relevant outputs for integrated solution	Supports common HealthConnect objective	Monitor readiness and unresolved dependencies	Upd



HealthConnect_Week6_Project_Su...

15 KB Word document

Project Management Track | Analytics and AI | Experience Lab | Interimship

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI**Central Question:** How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?**Week 6 Summary**

1. Planned: Integrate the HealthConnect workstreams, update dependencies and risks, validate integration readiness, document issues and decisions, and prepare coordination for Week 7 testing/refinement.

2. Completed: The PM integration controls were updated, and a meaningful Data Analytics -- Project Management integration was completed using the Week 5 Data Analytics analysis output.

3. Improved from Week 5: The work moved from task-level execution control to evidence-based integration control, including integration readiness, dependency tracking, issue/decision logging, and explicit Week 7 testing preparation.

4. Integrated: The Data Analytics Week 5 analysis was received and reviewed as an input to PM integration. The findings, limitations, and remaining analytical work were incorporated into PM coordination records.

5. Collaborated Track(s): Data Analytics. Other technical pathways remain pending evidence in the PM record.

6. Information Exchanged: Data Analytics provided its Week 5 appointment attendance/no-show analysis, including key findings, relevant factors/features, limitations, and remaining analysis. PM provided the integration context, dependencies, priorities, and Week 7 testing/refinement focus.

7. What Changed: The analytical findings are now represented as a documented PM integration input. Booking lead time and previous no-show history are treated as important analytical considerations; remaining analytical work and limitations are carried into Week 7 planning. No unverified downstream model use is claimed.

8. Findings / Outcomes: The received analysis identified booking lead time and previous no-show history as important factors associated with no-shows, while appointment type showed limited influence. The analysis also identified remaining multi-variable analysis and a fourth KPI as Week 6/next-stage work.



HealthConnect_Week6_Project_Int...

25 KB Word document



Felix has paused their notifications

Project

HealthConnect Clinic Experience Lab



Message Felix



Home



DMs



Activity



More



important result was the conversion of the Week 4 project foundation into actionable task control responsibility mapping, priorities, progress monitoring, risks, dependencies and coordination points.

testing/refinement.
2. Completed: The PM integration controls were updated, and a meaningful Data Analytics → Project Management integration was completed using the Week 5 Data Analytics analysis output.

3. Improved from Week 5: The work moved from task-level execution control to evidence-based integration control, including integration readiness, dependency tracking, issue/decision logging, and



Felix

3 tabs

...the Week 5 analysis was received and reviewed as an input to the PM integration controls, and remaining analytical work were incorporated into the



5. Collaborated Track(s): Data Analytics. Other technical pathways remain pending evidence in the PM record.

6. Information Exchanged: Data Analytics provided its Week 5 appointment attendance/no-show analysis, including key findings, relevant factors/features, limitations, and remaining analysis. PM provided the integration context, dependencies, priorities, and Week 7 testing/refinement focus.

7. What Changed: The analytical findings are now represented as a documented PM integration input. Booking lead time and previous no-show history are treated as important analytical considerations; remaining analytical work and limitations are carried into Week 7 planning. No unverified downstream model use is claimed.

8. Findings / Outcomes: The received analysis identified booking lead time and previous no-show history as important factors associated with no-shows, while appointment type showed limited influence. The analysis also identified remaining multi-variable analysis and a fourth KPI as Week 6/next-stage work.



HealthConnect_Week6_Project_Int...
 25 KB Word document

Programme	AnalystLab Africa Experience Lab Internship Programme
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Week	Week 6 — Integration, Advanced Development & Validation
Document Purpose	Project Integration & Coordination Package

Document Control

Version: 1.0

Issue Date: Week 6

Status: Working submission package — technical integration evidence remains subject to verification.

1. Week 5 – Week 6 Transition

The main Week 5 output was the HealthConnect Project Execution & Progress Management structure. Its most important result was the conversion of the Week 4 project foundation into actionable task control, responsibility mapping, priorities, progress monitoring, risks, dependencies and coordination points.

The main limitation carried into Week 6 was the lack of verified technical workstream status within the PM evidence set. During Week 6, a Data Analytics Week 5 analysis was subsequently received and reviewed, creating a verified Data Analytics → Project Management integration point. Other technical workstream outputs remain subject to verification.

Relevant workstreams are Data Analytics, Data Science, Machine Learning Engineering and Generative AI. The Week 6 improvement is to establish how these outputs connect, identify what is ready or still requires development, and manage the conditions required for broader testing.

2. Integration Objective

The Week 6 PM integration objective is to establish a clear, evidence-based view of how HealthConnect workstreams connect, which dependencies affect progress, which deliverables are ready for integration, which activities require further development, what issues and decisions require management attention, and what must be completed before Week 7 testing and refinement.

3. Workstream Integration Map